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Calculus & Trigonometry Reference Guide

Compiled Reference Document · Exponent Standard · Catppuccin Mocha Mauve

1 Calculus Working Methodology & Rules

- **Step 1: Global Constants Extraction** — Always factor out any overall constant multiplier from the integrand and keep it to the left of the integral symbol before evaluating or substituting.
- **Strict Exponential Notation for Variables** — Avoid radical signs ($\sqrt{}$) and variable denominators:
 - [label=o]
 - Use $x^{1/2}$ instead of radical forms.
 - Use $x^{-1/2}$ instead of fractions with roots in the denominator.
 - Use x^{-1} , x^{-2} , $(kx)^{-n}$ instead of division by variable expressions.
- **Fractions for Pure Constants Only** — Rational fraction bars are reserved strictly for constant values (e.g., $\frac{2}{5}$).
- **Explicit Step Cancellation** — Explicitly show the simplification of exponents during substitutions (e.g., $x^{-1/2} \cdot x^{1/2} = x^0 = 1$).

2 Trigonometric Derivatives & Integrals

2.1 Basic Trigonometric Functions

Function $f(x)$	Derivative $\frac{d}{dx}[f(x)]$	Indefinite Integral $\int f(x) dx$
$\sin(x)$	$\cos(x)$	$-\cos(x) + C$
$\cos(x)$	$-\sin(x)$	$\sin(x) + C$
$\tan(x)$	$\sec^2(x)$	$\ln \sec(x) + C = -\ln \cos(x) + C$
$\csc(x)$	$-\csc(x)\cot(x)$	$-\ln \csc(x) + \cot(x) + C$
$\sec(x)$	$\sec(x)\tan(x)$	$\ln \sec(x) + \tan(x) + C$
$\cot(x)$	$-\csc^2(x)$	$\ln \sin(x) + C$

2.2 Common Trigonometric Integral Forms & Cal 2 Powers

Integrand	Result
$\int \sec^2(x) dx$	$\tan(x) + C$
$\int \csc^2(x) dx$	$-\cot(x) + C$
$\int \sec(x)\tan(x) dx$	$\sec(x) + C$
$\int \csc(x)\cot(x) dx$	$-\csc(x) + C$
$\int \sin^2(x) dx$	$\frac{1}{2}x - \frac{1}{4}\sin(2x) + C$
$\int \cos^2(x) dx$	$\frac{1}{2}x + \frac{1}{4}\sin(2x) + C$
$\int \sec^3(x) dx$	$\frac{1}{2}\sec(x)\tan(x) + \frac{1}{2}\ln \sec(x) + \tan(x) + C$
$\int \csc^3(x) dx$	$-\frac{1}{2}\csc(x)\cot(x) - \frac{1}{2}\ln \csc(x) + \cot(x) + C$
$\int \tan^2(x) dx = \int (\sec^2(x) - 1) dx$	$\tan(x) - x + C$
$\int \cot^2(x) dx = \int (\csc^2(x) - 1) dx$	$1 - \cot(x) - x + C$

3.3 Power-Reducing Formulas (Half-Angle Integral Identities)

- $\sin^2(x) = \frac{1}{2}(1 - \cos(2x)) = \frac{1}{2} - \frac{1}{2} \cos(2x)$
- $\cos^2(x) = \frac{1}{2}(1 + \cos(2x)) = \frac{1}{2} + \frac{1}{2} \cos(2x)$
- $\tan^2(x) = (1 - \cos(2x)) \cdot (1 + \cos(2x))^{-1}$

4 Common Trigonometric Values (Unit Circle)

4.1 Special Angles (First Quadrant)

Deg	Rad	$\sin(\theta)$	$\cos(\theta)$	$\tan(\theta)$	$\csc(\theta)$	$\sec(\theta)$	$\cot(\theta)$	
0°	0	0	1	0	Undefined	1	Undefined	
30°	$\frac{1}{6}\pi$	$\frac{1}{2}$	$\frac{1}{2} \cdot 3^{1/2}$	$\frac{1}{3} \cdot 3^{1/2}$	2	$\frac{2}{3} \cdot 3^{1/2}$	$3^{1/2}$	
45°	$\frac{1}{4}\pi$	$\frac{1}{2} \cdot 2^{1/2}$	$\frac{1}{2} \cdot 2^{1/2}$	1	$2^{1/2}$	$2^{1/2}$	1	
60°	$\frac{1}{3}\pi$	$\frac{1}{2} \cdot 3^{1/2}$	$\frac{1}{2}$	$3^{1/2}$	$\frac{2}{3} \cdot 3^{1/2}$	2	$\frac{1}{3} \cdot 3^{1/2}$	
90°	$\frac{1}{2}\pi$	1	0	Undefined	1	Undefined	0	

4.2 Quadrantal Angles & Quadrant Signs (ASTC Rule)

- $\theta = 0 \implies (x, y) = (1, 0), \tan(0) = 0$
- $\theta = \frac{1}{2}\pi \implies (x, y) = (0, 1), \tan(\frac{1}{2}\pi)$ is Undefined
- $\theta = \pi \implies (x, y) = (-1, 0), \tan(\pi) = 0$
- $\theta = \frac{3}{2}\pi \implies (x, y) = (0, -1), \tan(\frac{3}{2}\pi)$ is Undefined
- Quadrant I ($0 < \theta < \frac{1}{2}\pi$): All positive (+)
- Quadrant II ($\frac{1}{2}\pi < \theta < \pi$): Sine and Csc positive (+)
- Quadrant III ($\pi < \theta < \frac{3}{2}\pi$): Tangent and Cot positive (+)
- Quadrant IV ($\frac{3}{2}\pi < \theta < 2\pi$): Cosine and Sec positive (+)

5 Exponential & Logarithmic Functions

5.1 Derivatives

Function $f(x)$	Derivative $f'(x)$	Chain Rule Form $\frac{d}{dx}[f(u)]$
e^x	e^x	$u' \cdot e^u$
a^x ($a > 0, a \neq 1$)	$a^x \ln(a)$	$u' \cdot a^u \ln(a)$
$\ln(x)$	x^{-1}	$u' \cdot u^{-1}$
$\log_a(x)$	$(\ln a)^{-1} \cdot x^{-1}$	$u' \cdot (\ln a)^{-1} \cdot u^{-1}$

5.2 Integrals

Integrand	Indefinite Integral	Technique / Note
$\int e^x dx$	$e^x + C$	Standard
$\int e^{kx} dx$	$\frac{1}{k} e^{kx} + C$	u -substitution
$\int a^x dx$	$(\ln a)^{-1} a^x + C$	Exponential base a
$\int x^{-1} dx$	$\ln x + C$	Power -1 rule
$\int \ln(x) dx$	$x \ln(x) - x + C$	Parts: $u = \ln x, dv = dx$
$\int \log_a(x) dx$	$(\ln a)^{-1} (x \ln x - x) + C$	Change of base

5.3 Common Integration by Parts Combinations

- $\int x e^x dx = e^x(x - 1) + C$
- $\int x^2 e^x dx = e^x(x^2 - 2x + 2) + C$
- $\int x \ln(x) dx = \frac{1}{2}x^2 \ln(x) - \frac{1}{4}x^2 + C$
- $\int e^x \sin(x) dx = \frac{1}{2}e^x(\sin x - \cos x) + C$
- $\int e^x \cos(x) dx = \frac{1}{2}e^x(\cos x + \sin x) + C$

6 Inverse Trigonometric Functions (Arcs)

6.1 Derivatives of Inverse Trig Functions

Function $f(x)$	Derivative $f'(x)$	Chain Rule Form $\frac{d}{dx}[f(u)]$
$\arcsin(x)$	$(1 - x^2)^{-1/2}$	$u' \cdot (1 - u^2)^{-1/2}$
$\arccos(x)$	$-(1 - x^2)^{-1/2}$	$-u' \cdot (1 - u^2)^{-1/2}$
$\arctan(x)$	$(1 + x^2)^{-1}$	$u' \cdot (1 + u^2)^{-1}$
$\operatorname{arccot}(x)$	$-(1 + x^2)^{-1}$	$-u' \cdot (1 + u^2)^{-1}$
$\operatorname{arcsec}(x)$	$ x ^{-1}(x^2 - 1)^{-1/2}$	$u' \cdot u ^{-1}(u^2 - 1)^{-1/2}$
$\operatorname{arccsc}(x)$	$- x ^{-1}(x^2 - 1)^{-1/2}$	$-u' \cdot u ^{-1}(u^2 - 1)^{-1/2}$

6.2 Integrals Resulting in Inverse Trig Functions

For constant $a > 0$:

- $\int (a^2 - x^2)^{-1/2} dx = \arcsin\left(\frac{1}{a}x\right) + C$
- $\int (a^2 + x^2)^{-1} dx = \frac{1}{a} \arctan\left(\frac{1}{a}x\right) + C$
- $\int |x|^{-1}(x^2 - a^2)^{-1/2} dx = \frac{1}{a} \operatorname{arcsec}\left(\frac{1}{a}|x|\right) + C$

6.3 Integrals of Inverse Trig Functions (Integration by Parts)

- $\int \arcsin(x) dx = x \arcsin(x) + (1 - x^2)^{1/2} + C$
- $\int \arccos(x) dx = x \arccos(x) - (1 - x^2)^{1/2} + C$
- $\int \arctan(x) dx = x \arctan(x) - \frac{1}{2} \ln(1 + x^2) + C$
- $\int \operatorname{arccot}(x) dx = x \operatorname{arccot}(x) + \frac{1}{2} \ln(1 + x^2) + C$
- $\int \operatorname{arcsec}(x) dx = x \operatorname{arcsec}(x) - \ln|x + x(1 - x^{-2})^{1/2}| + C$
- $\int \operatorname{arccsc}(x) dx = x \operatorname{arccsc}(x) + \ln|x + x(1 - x^{-2})^{1/2}| + C$